

Read the passage below, making note of any important points and definitions.

Scientists go about their work, gathering and interpreting data in order to better understand nature. Rarely does a scientist question the basis for this work. What do we mean by “better understand”? How do we know our methods are valid and our interpretations correct? Why should we even assume that nature can be understood at all? These are questions of a type that scientists rarely ask. Asking this kind of question is the task of the philosopher of science. Philosophers of science have taken a variety of approaches. For some, the goal has been to ascertain the true nature of things, which the methods of science can only hint at. This metaphysical outlook has come under strong attack by other philosophers, who claim that only direct observations have meaning and that the job of science is to find regularities in these observations. In this view, the role of the philosopher is a sort of gatekeeper, to bar

metaphysical entities from science. One school of thought assigns to philosophy the work of determining normative methodological rules for the sciences. In other words, philosophers need to tell scientists how they ought to judge between competing theories; select valid evidence; construct explanations with desirable properties; use inductive and deductive logic; determine the truth or falsehood of a claim; and so on. This view was predominant during the first half of the twentieth century, but has since been challenged (primarily by thinkers influenced by the history of science). Historical studies indicate that scientists rarely behave according to the rules prescribed by the philosophers, so what meaning do such rules really have? In this chapter, we’ll take a brief look at these controversies, along with a variety of other interesting issues, acquiring a broad overview of the philosophical landscape as we go.

Gregory N Derry. What science is and how it works. Princeton University Press, 1999.

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CAUSALITY AND DETERMINISM The idea of causality is very important in the sciences but has been notoriously difficult to make philosophically precise. The root word of causality is “cause.” If one event or action is the cause of another, they are connected by a causality relation. This sounds simple enough, but what do we mean when we say that one thing causes another? For example, what caused my coffee cup to break? I pushed my coffee cup off the shelf. So, my pushing it caused the cup to break. But we could just as well say that gravity pulling it down caused the cup to break; or, we could say that the brittleness of the clay caused the cup to break. This example illustrates two problems: the problem of multiple partial causes, and the problem of logical necessity (i.e., if the cause occurs, the effect must follow). In the physical sciences, discussions of causality are restricted to cases where a single cause can be identified that always gives rise to a well-defined effect. In the life sciences and the social sciences, where such a restriction is seldom possible, the concept of causality is still often employed (e.g., a virus causes the common cold, but exposure to the virus doesn’t always necessarily result in a cold). Although descriptions based on causality are quite widespread and important in these fields, philosophical analysis of such usage has not been thoroughly worked out. The restriction to cases of single causes and necessary effects is very limiting, of course, but does permit a high degree of logical rigor in the discussion.

Empiricist philosophers, however, have used this criterion of universal conjunction to deny the meaningfulness of causality. As David Hume pointed out, the most that we can ever really know is that one event has always followed another; to then claim that the first event causes the second is merely a bad verbal habit. Night always follows day, for example, but we don’t claim that the day causes the night. This isn’t the last word, though, because empiricism isn’t the only game in town; other philosophers have elevated causality to the status of a metaphysical principle. Philosophers who were heavily influenced by physics adopted a different line of thought. Since you can write down equations and solve them in order to predict the second event based on the occurrence of the first event, then predictability becomes the key issue. Another word for such predictability is determinism; the first event determines the occurrence of the second. Put differently, the important property of the world is that it’s deterministic, not that it’s causal. Causality, in

this view, is just anthropomorphic language assigning motives to a deterministic sequence.

THE QUINE-DUHEM THESIS In our discussion of stellar parallax, we saw that an apparently disconfirming fact could be accounted for by the overall context of the situation (large distances, limited instrumentation, etc.). This episode is indicative of a more general problem, namely the underdetermination of theories by facts. The general idea is this: Any scientific theory is a complex interlocking set of concepts, observations, definitions, presuppositions, experimental results, and connections to other theories. No single fact is going to be crucial to the survival of a theory. We can illustrate this point with a simple hypothetical example. Suppose I have a theory that predicts that carbon turns to diamond at a certain very high pressure and temperature. I try it, and it doesn’t happen. My theory appears to be falsified. But, how do I know what the temperature and pressure really are? To measure extreme pressures and temperatures requires special instruments, which themselves operate according to theories. Maybe those theories are false under our conditions, and my diamond theory is correct after all. Empirical facts are not always unambiguous. So, the complexity of many interdependent and theory-dependent terms makes it difficult to frame an unambiguously falsifiable statement. Pierre Duhem argued as early as 1906 that our experiments only test the total structure of a science, not individual hypotheses. W. V. O. Quine later extended and strengthened this idea considerably, resulting in the so-called Quine-Duhem thesis: It’s impossible, on purely logical grounds, to falsify a theory experimentally, because any observation can be accommodated by making suitable adjustments.

The Quine-Duhem thesis has generated a lot of controversy. Some thinkers believe the claim is a triviality, while others believe that it utterly destroys the objectivity of science. An important point to keep in mind is that the adjustments we must make to save a theory, while they may be logically possible, may also be unacceptable on other legitimate grounds; the Quine-Duhem thesis does not mean that all theories are equally good. The question then becomes this: On what grounds are such judgments made?

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